Automatic Bloom's Taxonomy Question Generation Using RAG and ModernBERT Classification with Regenerative Feedback

Faishal Fariz Hidayatullah, Dwi Sunaryono

***Abstract*— Automatic question generation has emerged as a key enabler of scalable educational assessment, yet existing approaches cannot reliably ensure that generated questions align with the intended cognitive levels of Bloom's Taxonomy, often relying on coarse difficulty labels rather than explicit cognitive-level mapping. This study presents a framework integrating Retrieval-Augmented Generation (RAG), four open-weight Large Language Models, and a pretrained ModernBERT Bloom's Taxonomy classifier with regenerative feedback. RAG retrieves relevant passages from a curated OpenStax corpus, the LLMs generate questions targeting Bloom's levels from Remember (C1) to Create (C6), and ModernBERT verifies each question's cognitive level. When predictions differ from the target level, the framework regenerates the question iteratively. Llama-3.1-8B, Qwen-3-8B, Mistral-7B, and Gemma-3-4B were evaluated under identical 4-bit precision via Ollama on 1,914 generated questions per model. ModernBERT achieved 86.36% accuracy on the Kaggle Bloom's Taxonomy dataset. Because the end-to-end evaluation relies on this same classifier for verification, the resulting metric reflects pipeline self-consistency rather than externally validated pedagogical correctness. Under this metric, Llama-3.1-8B achieved the highest classifier-verified alignment rate (80.93%) and overall F1-score (0.8946), leading at four of six Bloom's levels, while Gemma-3-4B ranked second. On a single consumer-grade GPU, the framework generated 1,325–3,499 cognitively verified questions per hour. These findings show that classifier-guided regenerative feedback improves cognitive-level alignment while supporting scalable automatic question generation.**

***Index Terms*—Automatic Question Generation, Bloom's Taxonomy, Large Language Models, ModernBERT, Retrieval-Augmented Generation.**

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I. INTRODUCTION

THE development of artificial intelligence (AI) technology has brought significant changes in the field of education, particularly in the development of digital-based learning evaluation systems [1] [2]. One of the main challenges in the learning process is the preparation of questions that are not only relevant to the material but also appropriate to the cognitive abilities of the learners. Bloom's Taxonomy is a framework commonly used to classify levels of thinking ability, ranging from lower-order thinking skills such as Remember (C1) to higher-order thinking skills such as Create (C6) [3]. Ensuring that generated questions are appropriately mapped to these six cognitive levels remains a non-trivial challenge, particularly when done automatically at scale.

The emergence of Large Language Models (LLMs) and transformer-based architectures has opened new possibilities for automating educational content generation. However, LLMs alone tend to produce questions without guaranteed alignment to a specific cognitive level, as they lack an explicit verification mechanism to ensure the generated output conforms to a defined Bloom's Taxonomy level [4]. This limitation motivates the integration of a dedicated classifier to provide post-generation cognitive-level verification.

Previous research employs Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs) to automatically generate questions from knowledge bases [5]. The results indicate that models such as Llama-3-8B, GPT-2, and Cohere can produce questions with high accuracy, but still face limitations in maintaining consistent difficulty levels and stable question quality, and do not utilize Bloom's Taxonomy to classify question difficulty. Other studies have explored the application of machine learning classifiers to categorize questions according to Bloom's Taxonomy levels, yet these approaches operate independently of a generation pipeline and rely on fixed, manually curated question banks [6].

Despite their contributions, prior studies remain limited because the first focuses only on cognitive classification without generating new questions, while the second generates questions but does not align difficulty with Bloom's Taxonomy C1–C6. In addition, RAG-based studies generally apply only broad difficulty categories such as low, medium, and high without explicit cognitive mapping.

A critical component that addresses the cognitive classification gap is the availability of pre-trained transformer-based classifiers fine-tuned specifically for Bloom's Taxonomy. This study leverages ModernBERT-Bloom-Taxonomy [7], a publicly available model on Hugging Face (manzarimalik/ModernBERT-Bloom-Taxonomy) that was fine-tuned from answerdotai/ModernBERT-large on a structured Bloom's Taxonomy dataset [8]. ModernBERT-large itself represents a significant modernization of the classical BERT encoder architecture, introducing Rotary Positional Embeddings (RoPE) for extended sequence support up to 8,192 tokens, alternating local-global attention mechanisms for efficient long-context processing, GeGLU activations, and Flash Attention with unpadding for hardware-optimized inference [9]. Pre-trained on two trillion tokens of English and code data, ModernBERT provides a substantially stronger representational foundation compared to its predecessor, making it well-suited for fine-grained text classification tasks such as cognitive-level prediction across the six Bloom's Taxonomy levels (BT1–BT6). The use of this pre-trained classifier eliminates the need to train a Bloom's classifier from scratch and provides a reliable, reproducible cognitive verification component within the proposed pipeline.

This study proposes an AI-based framework integrating Retrieval-Augmented Generation (RAG), four open-weight Large Language Models (LLMs), and the ModernBERT-Bloom-Taxonomy pretrained classifier to automatically generate questions with verified cognitive levels based on Bloom's Taxonomy. This integrated approach enables the production of contextually relevant and cognitively verifiable assessment items, supporting adaptive learning in Learning Management Systems (LMS). Accordingly, the contributions of this study include:

- Integration of RAG, LLMs, and a pre-trained ModernBERT-based classifier for automated question generation with cognitive-level verification.
- Explicit classification of question difficulty using Bloom’s Taxonomy C1–C6 instead of general difficulty labels
- A regenerative feedback mechanism that iteratively refines generated questions to achieve cognitive-level alignment.
- Comprehensive comparative evaluation of four open-weight LLMs on cognitive-level alignment, providing empirical guidance on model selection for Bloom's Taxonomy question generation.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed framework and methodology. Section IV reports the experimental setup and evaluation results. Section V concludes the paper and discusses future work.

II. RELATED WORK

This section reviews three interconnected research streams that form the foundation of the proposed framework: (A) automatic question generation using Large Language Models and RAG-based approaches, (B) Bloom's Taxonomy cognitive-level classification, and (C) transformer-based encoder models for educational text classification. A summary of the reviewed studies and their respective capabilities is presented in Table I, which serves as the basis for identifying the research gap addressed by this work.

A. Automatic Question Generation With Rag

Automatic Question Generation (AQG) has been a long-standing challenge in educational AI, evolving from early rule-based and template-driven approaches toward deep learning and generative models [10]. With the emergence of Large Language Models (LLMs), AQG systems can now produce contextually coherent questions from unstructured text with minimal human supervision.

Aisyah et al. [5] conducted a comparative study on RAG-based question generation using multiple LLMs including Llama-3-8B, GPT-2, and Cohere, evaluating their performance in generating questions from domain-specific knowledge bases. Their system retrieved relevant passages using vector similarity search and prompted the LLMs to generate questions accordingly. The study demonstrated that RAG substantially reduces hallucinations compared to standalone LLM generation, and that Llama-3-8B achieved competitive performance in terms of relevance and fluency. However, the authors noted persistent limitations in maintaining consistent difficulty levels across generated questions. Critically, no mechanism was employed to classify or verify the cognitive complexity of generated outputs against an established educational taxonomy

Maity et al. [11] explored In-Context Learning (ICL), RAG, and a hybrid RAG-ICL approach for AQG using GPT-4 and BART. Their hybrid model consistently outperformed individual methods in contextual accuracy and question relevance. Despite this, the study relied solely on automated metrics such as ROUGE and BLEU for evaluation, without assessing alignment to any cognitive framework. The absence of explicit cognitive-level targeting limits the pedagogical utility of the generated questions, particularly for structured assessments in Learning Management Systems (LMS).

Scaria et al. [12] examined the ability of five instruction fine-tuned LLMs including GPT-4, Mistral, and Llama variants to generate educational questions at different Bloom's Taxonomy levels using advanced prompting strategies. Their findings confirmed that LLMs can generate high-quality questions when prompted with adequate cognitive-level information, but exhibited significant variance in skill alignment, particularly for higher-order cognitive levels (C4–C6). The study also showed that automated evaluation metrics are insufficient for verifying cognitive alignment, highlighting the need for a dedicated classifier as a post-generation

TABLE I
COMPARISON OF RELATED STUDIES

Study	Method	Question Generation	Bloom's Taxonomy	Cognitive Verification	Feedback Loop	RAG
Aisyah et al. [5] (2024)	RAG + Llama-3-8B / GPT-2 / Cohere	✓	✗	✗	✗	✓
Banujan et al. [6] (2023)	ANN + LSTM + BERT	✗	✓ (prompt-guided)	✓ (classifier only)	✗	✗
Maity et al. [11] (2025)	RAG + ICL + GPT-4 / BART	✓	✗	✗	✗	✓
Scaria et al. [12] (2024)	Prompting + LLMs (GPT-4, Mistral, Llama)	✓	✓ (prompt-guided)	✗ (no classifier)	✗	✗
Kumar et al. [13] (2025)	ML / RNN / BERT / RoBERTa / LLM	✗	✓ (C1–C6)	✓ (classifier only)	✗	✗
Warner et al. [9] (2024)	ModernBERT pre-training	✗	✗	✗	✗	✗
Proposed Work	RAG + 4 LLM Models + ModernBERT-Bloom	✓	✓ (C1–C6)	✓ (classifier)	✓	✓

verification mechanism.

Collectively, these studies demonstrate the strengths of RAG in grounding question generation to relevant content while exposing a shared limitation: the absence of a closed-loop mechanism to verify and enforce cognitive-level alignment. The present work directly addresses this gap by integrating a pre-trained Bloom's Taxonomy classifier into the generation pipeline and incorporating a regenerative feedback loop that iteratively refines non-ideal outputs.

B. Bloom's Taxonomy Classification

Parallel to generation research, a distinct body of work focuses on the automatic classification of existing questions according to Bloom's Taxonomy cognitive levels. These studies treat classification as a standalone task applied to pre-existing question banks, without coupling it to a generation component.

Banujan et al. [6] developed an automated classification system for undergraduate examination questions using deep learning models, specifically Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks combined with word embedding techniques including GloVe, BERT, and TF-IDF. Their study collected 16,584 questions from multiple Sri Lankan universities and applied the Revised Bloom's Taxonomy for labeling. Results showed that the LSTM+BERT model significantly outperformed ANN+TF-IDF and LSTM+GloVe configurations based on ANOVA with Bonferroni correction, demonstrating the superiority of contextual embeddings for cognitive-level prediction. However, the system operates entirely on existing, manually curated question banks and provides no mechanism for generating new questions. The classification component is therefore decoupled from the educational content retrieval process.

Kumar et al. [13] conducted a systematic comparison of classification paradigms for Bloom's Taxonomy across Naive Bayes, Logistic Regression, SVM, LSTM, BiLSTM, GRU, BERT, RoBERTa, and LLM-based zero/few-shot approaches on a dataset of 600 labeled sentences. Their analysis identified research gaps in the existing literature: first, most studies rely on conventional ML with surface-level features; second, the effectiveness of deep learning on small datasets remains underexplored; and third, LLM-based zero-shot classification introduces new opportunities not yet systematically benchmarked. While their work provides a comprehensive benchmark, it confirms that classification-only approaches do not address the need for integrated cognitive-level question generation.

These studies collectively establish that transformer-based classifiers, particularly BERT-family models, represent the current state of the art in Bloom's Taxonomy classification. However, they operate independently of question generation pipelines and do not provide a mechanism for producing new assessment content aligned to target cognitive levels.

C. Transformer-Based Encoders

The backbone of the cognitive-level classifier in the proposed framework is ModernBERT, a recently introduced encoder-only transformer that represents a significant architectural advancement over classical BERT models [14]. Warner et al. [9] introduced ModernBERT as a Pareto improvement over prior encoders, trained on 2 trillion tokens of English and code data with a native context length of up to 8,192 tokens. The architecture incorporates Rotary Positional Embeddings (RoPE) for extended sequence support, alternating local-global attention patterns for efficient long-context processing, GeGLU activations, and Flash Attention with input unpadding for hardware-optimized inference. ModernBERT achieves state-of-the-art results on diverse classification and retrieval benchmarks, outperforming both base and large variants of RoBERTa, DeBERTa, and prior BERT checkpoints while maintaining significantly lower memory consumption and faster inference speeds.

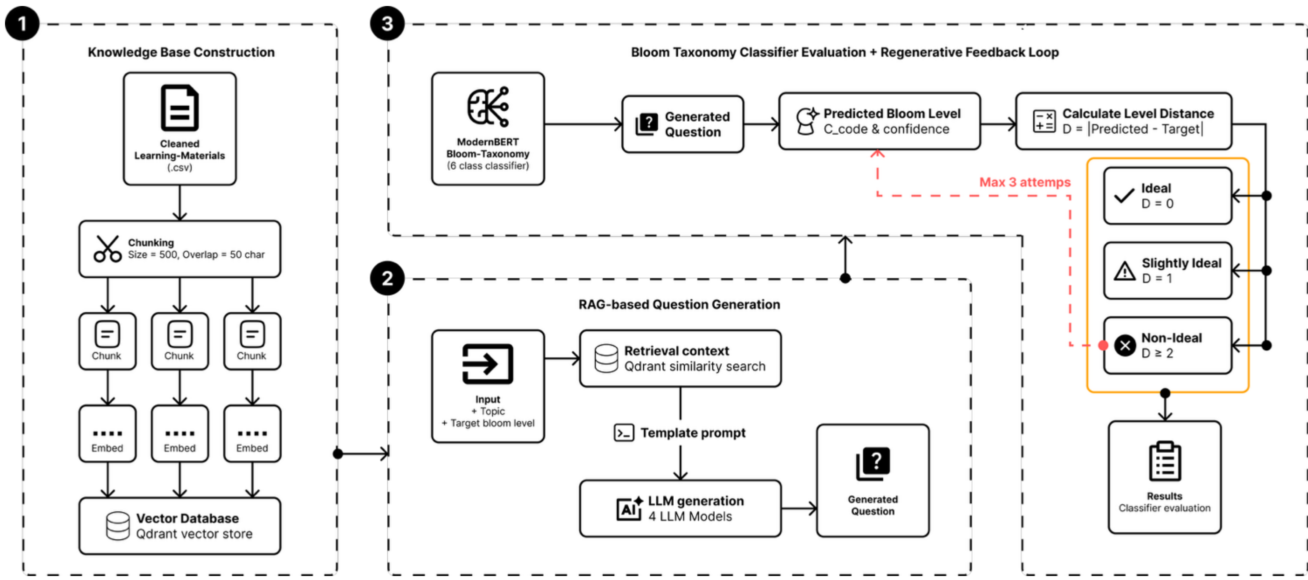

Fig. 1. Overview of the proposed three-phase pipeline.

Building on this foundation, Malik [7] fine-tuned ModernBERT-large on a structured Bloom's Taxonomy dataset (manzarimalik/ModernBERT-Bloom-Taxonomy, Hugging Face), yielding a six-class text classifier mapping educational questions to BT1 (Remember) through BT6 (Create). The model was fine-tuned using the Unsloth training framework and Hugging Face's TRL library, achieving 2x faster fine-tuning compared to standard procedures. Released under an Apache 2.0 license, this model provides a plug-and-play cognitive verification component that eliminates the need to train a Bloom's classifier from scratch. The availability of this pre-trained model is a key enabler of the proposed framework, as it allows cognitive-level verification to be performed efficiently within a real-time generation-evaluation loop.

III. Proposed Method

The proposed framework integrates three core components: Retrieval-Augmented Generation (RAG), multiple Large Language Models (LLMs), and a pre-trained Bloom's Taxonomy classifier into a unified pipeline for automated cognitive-level question generation. As illustrated in Fig. 1, the system operates across three sequential phases.

A. Dataset

This study utilizes two distinct datasets, each serving a different purpose within the proposed framework. The first is a knowledge base dataset constructed from OpenStax learning materials, used to support the RAG-based question generation pipeline [15]. The second is a Bloom's Taxonomy classification dataset sourced from Kaggle, used to evaluate the cognitive-level classification performance of the pre-trained ModernBERT-Bloom-Taxonomy classifier [8].

OpenStax was selected as the data source due to its structured, academically rigorous content spanning multiple disciplines, its free availability, and its permissive license for research use. Five subjects are included in the dataset: Biology, Physics, Chemistry, Psychology, and Economics. Learning materials are extracted from the corresponding OpenStax textbook PDF for each subject, producing a four-column structured dataset with the fields content, topic, subject, and keywords. Table II presents the top five entries of the preprocessed OpenStax dataset as a representative sample of the knowledge base structure.

In addition to the OpenStax knowledge base, this study employs a publicly available Bloom's Taxonomy classification dataset. The dataset consists of 8,767 labeled educational questions spanning six cognitive levels defined by Bloom's Taxonomy: BT1 (Remember), BT2 (Understand), BT3 (Apply), BT4 (Analyze), BT5 (Evaluate), and BT6 (Create). Each entry contains two fields: a Questions column containing the question text, and a Category column containing the corresponding Bloom's Taxonomy level label. The questions originate from diverse academic domains and vary in length and linguistic complexity across cognitive levels. Table III presents the top five entries of the dataset as a representative sample of its structure.

B. Preprocessing

The OpenStax knowledge base dataset undergoes a two-stage preprocessing pipeline: content extraction and text cleaning, as illustrated in Fig. 2. The goal of this pipeline is to transform raw PDF-extracted content into clean, structured text suitable for embedding and retrieval in the RAG pipeline.

In the content extraction stage, each OpenStax textbook PDF is downloaded programmatically and parsed using the document's TOC and bookmark hierarchy. Sections at depth levels 1–4 are retained as candidate entries. Each section is subject to quality filtering: entries with fewer than 300 characters are discarded, entries exceeding 5,000 characters are truncated at the last complete sentence, and non-content sections such as Exercises, Glossary, and References are

TABLE II
PREPROCESSED OPENSTAX KNOWLEDGE BASE DATASET

No	Content	Topic	Subject	Keywords
1	Genetics is the study of	Introduction to Patterns of Inheritance	Biology	introduction chapter…
2	Today, members of this plant …	Speciation	Biology	species, supercontinent…
3	In both prokaryotes ….	Transcription	Biology	dna, learning objectives …
4	Another scenario in which ….	Evidence of Evolution	Biology	population, founder effect …
5	Like terrestrial biomes …	Aquatic and Marine Biomes	Biology	wildlife refuge, national wildlife ….

TABLE III
BLOOM'S TAXONOMY CLASSIFICATION DATASET

Questions	Category
About what proportion of the population of the US is living on farms?	BT 1
Correctly label the brain lobes indicated on the diagram below.	BT 1
Define compound interest.	BT 1
Define four types of traceability.	BT 1
Define mercantilism.	BT 1

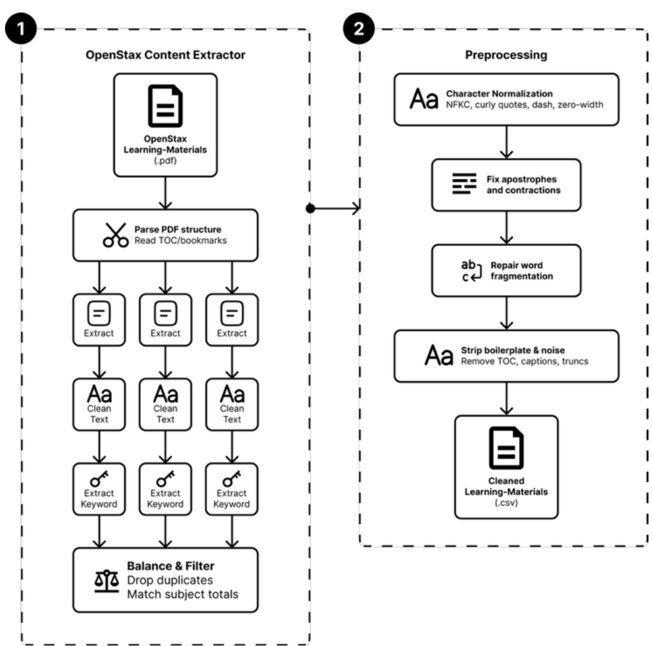

Fig. 2. OpenStax two-stage preprocessing pipeline.

removed. Keywords are extracted using YAKE [16], retaining the top five bigram keywords per section. A stratified sampling procedure is then applied to equalize the number of entries across all five subjects, with duplicates removed prior to sampling.

Following extraction, the dataset undergoes a six-step text cleaning pipeline to address PDF extraction artefacts. Step 1 normalizes characters to NFKC Unicode form, standardizes punctuation marks, removes zero-width characters, and collapses whitespace. Step 2 restores apostrophes dropped during extraction using rule-based pattern matching for contractions and possessives. Step 3 repairs word fragmentation artefacts caused by PDF multi-column layout using a predefined correction dictionary. Step 4 removes boilerplate TOC prefixes, trailing chapter outline suffixes, and image credit annotations. Step 5 removes inline figure captions, contextual section markers, and figure references. Step 6 repairs entries truncated at the character boundary by trimming to the last complete sentence, provided the terminal punctuation occurs beyond 50% of the total text length.

C. Knowledge Base Construction

The preprocessed OpenStax dataset is loaded as a collection of LangChain Document objects, with each row's content field serving as the document text and the topic, subject, and keywords fields preserved as metadata. Each document is subsequently split into fixed-size chunks using a RecursiveCharacterTextSplitter with a chunk size of 500 characters and an overlap of 50 characters, using the separator hierarchy ["\n\n", "\n", ". ", " ", ""] to ensure splits occur at natural linguistic boundaries where possible. Topic and subject metadata are propagated to each chunk to support filtered retrieval.

Each chunk is encoded into a 384-dimensional dense vector using the all-MiniLM-L6-v2 sentence embedding model [17], a lightweight transformer-based encoder pre-trained on large-scale sentence-pair datasets for semantic similarity tasks. The embedding of a chunk d is defined as Equation (1).

$$e(d) = Encoder(d), e(d) \in \mathbb{R}^{384} \quad (1)$$

All embeddings are L2-normalized, such that $\|e(d)\| = 1$. The encoded vectors are stored in an in-memory Qdrant vector database [18], configured with cosine distance as the similarity metric. Cosine similarity between a query vector $e(q)$ and a document vector $e(d)$ is computed as Equation (2).

$$sim(q,d) = \frac{(e(q) \cdot e(d))}{(\|e(q)\| \cdot \|e(d)\|)} \quad (2)$$

Since all vectors are L2-normalized, the cosine similarity computation reduces to the inner product $e(q) \cdot e(d)$. Given an input query q, the retriever returns the three document chunks with the highest semantic similarity scores from the vector database. These retrieved chunks are subsequently used as contextual information for the RAG-based question generation process. The resulting retriever R forms the document supply component of the RAG pipeline described in Section E.

D. Bloom's Taxonomy Classifier

The cognitive-level classifier used in this study is ModernBERT-Bloom-Taxonomy, a publicly available model on Hugging Face fine-tuned from answerdotai/ModernBERT-large on the Bloom's Taxonomy classification dataset described in Section 3.2. ModernBERT-large is an encoder-only transformer pre-trained on two trillion tokens of English and code data with a native context length of 8,192 tokens. Its architecture incorporates Rotary Positional Embeddings (RoPE) for extended sequence support, alternating local-global attention for efficient long-context processing, GeGLU activations, and Flash Attention with input unpadding for hardware-optimized inference [9].

The fine-tuned model performs six-class text classification, mapping an input question to one of the Bloom's Taxonomy cognitive levels. The label mapping is defined in Table IV.

TABLE IV
BLOOM'S TAXONOMY LABEL MAPPING

Label	Cognitive Level	C-Code
BT1	Remember	C1
BT2	Understand	C2
BT3	Apply	C3
BT4	Analyze	C4
BT5	Evaluate	C5
BT6	Create	C6

The classifier is loaded as a HuggingFace text-classification pipeline and operates on the contextual representation of the [CLS] token produced by the ModernBERT encoder. Given an input question text x, the model produces a hidden representation h_[CLS] ∈ $\mathbb{R}^d$, which is passed through a linear classification head followed by a softmax activation as Equation (3).

$$P_{(c|x)} = softmax\big(W \cdot h_{[CLS]} + b\big), \quad c \in \{BT1, BT2, BT3, BT4, BT5, BT6\} \qquad (3)$$

Where W ∈ $\mathbb{R}^6$ is the weight matrix and b ∈ $\mathbb{R}^6$ is the bias vector of the classification head. The predicted Bloom's Taxonomy level is then defined as Equation (4).

$$\hat{y} = argmax_c P_{(c\,|\,x)} \qquad (4)$$

The classifier also returns the confidence score of the predicted class, defined as Equation (5).

$$conf(\hat{y}) = max_c P_{(c\,|\,x)} \qquad (5)$$

This model is loaded once at system initialization and reused across all classification calls in Phase 3, eliminating the overhead of repeated model loading during the feedback loop.

E. RAG-Based Question Generation

Question generation is performed by a RAG chain that combines the Qdrant retriever constructed in Phase 1A with a Large Language Model (LLM). RAG is an AI technique that enhances large language models by retrieving relevant information from external knowledge sources before generating a response [19]. The selected models, Llama-3.1-8B, Mistral-7B, Qwen-3-8B, and Gemma-3-4B, represent prominent open-weight LLM families with competitive performance in instruction-following, reasoning, and language understanding tasks, while remaining suitable for local deployment on consumer-grade hardware [20], [21]. This selection enables a comparative evaluation of cognitive-level question generation across models developed by different organizations. All models were deployed locally using Ollama [22] and executed on an NVIDIA GeForce RTX 3060 GPU with a temperature setting of 0.7.

The generation process takes two inputs: a topic τ and a target Bloom's Taxonomy level $L* \in$ {*Remember, Understand, Apply, Analyze, Evaluate, Create}*. The retriever is invoked with the topic as the query to obtain the *top-k = 3* most relevant chunks from the Qdrant vector store. The retrieved chunks are concatenated into a single context string *C*, this computed as Equation (6).

$$C = d_1 \oplus d_2 \oplus d_3, \quad where\, d_i \in R(\tau) \qquad (6)$$

Where $\oplus$ denotes string concatenation separated by a section delimiter. The context *C*, topic τ, and target level $L*$ are injected into a structured prompt template. The prompt is designed to guide the LLM in generating questions whose cognitive complexity strictly conforms to the target Bloom's level. To support reproducibility, the complete verbatim prompt template used in all experiments, including the inline Bloom's-level definitions, the full set of generation rules, and the exact output-formatting instructions, is presented in Table V.

To enforce cognitive-level specificity, the prompt supplies a distinct set of action verbs for each Bloom's level as generation constraints, as summarized in Table VI. The action verbs used as generation constraints are adapted from the revised Bloom's Taxonomy framework and commonly used Bloom verb inventories in educational assessment design [23].

Based on the prompt configuration and cognitive-level constraints described above, the question generation process is formalized in Equation (7).

$$Q = LLM_{(prompt(C,\tau,L*))}, \quad temp = 0.7 \qquad (7)$$

The LLM output is parsed by a string output parser that extracts individual question strings from the raw text, retaining only lines that contain a question mark as valid question candidates. The generated question Q is subsequently passed to Phase 3 for cognitive-level verification and feedback evaluation.

TABLE V
VERBATIM PROMPT TEMPLATE FOR QUESTION GENERATION

LLM Prompt
You are an expert educational-assessment designer. Generate exam questions whose cognitive complexity matches a target level of Bloom's Taxonomy (Revised). # Bloom's Taxonomy levels (use the listed action verbs) - C1 Remember — recall facts: define, list, identify, name, recall, state - C2 Understand — explain ideas: describe, explain, summarize, classify, compare - C3 Apply — use in new context: apply, demonstrate, solve, use, compute, illustrate - C4 Analyze — break apart ideas: analyze, contrast, examine, differentiate, infer - C5 Evaluate — justify a decision: assess, critique, judge, defend, justify, evaluate - C6 Create — produce new work: design, construct, develop, formulate, propose, plan # Retrieved learning-material context {C} # Task Topic: {τ} Target Bloom level: {$L*$} Number of questions: {num_questions} # Rules 1. Each question MUST clearly target the {$L*$} level (use its action verbs). 2. The question must be answerable from the context above (no hallucination). 3. Output one question per line. No numbering, no bullets, no commentary. 4. Do not write the answer — only the question.

TABLE VI
ACTION VERBS FOR EACH BLOOM'S LEVEL

Bloom's Level	Action Verbs
C1	define, list, identify, name, recall, state
C2	describe, explain, summarize, classify, compare
C3	apply, demonstrate, solve, use, compute, illustrate
C4	analyze, contrast, examine, differentiate, infer
C5	assess, critique, judge, defend, justify, evaluate
C6	design, construct, develop, formulate, propose, plan

F. Regenerative Feedback Loop

Upon generation of a candidate question Q from Phase 2, the ModernBERT-Bloom-Taxonomy classifier and is applied to predict the cognitive level of the generated question. The predicted level $\hat{L}$ is obtained as shown in Equation (8).

$$L^\wedge = argmax\_c\, P(c \mid Q) \tag{8}$$

The cognitive distance δ between the predicted level $L^\wedge$ and the target level $L*$ is then computed as the absolute difference between their ordinal positions in the Bloom's Taxonomy hierarchy, as shown in Equation (9).

$$\delta = |rank(L^\wedge) - rank(L*)| \tag{9}$$

Based on the value of δ, the generated question is assigned one of three status categories, as defined in Table VII. Questions with status Ideal or Slightly Ideal are accepted as the final output and recorded along with their predicted level, confidence score, and number of attempts used. The acceptance of $\delta = 1$ (Slightly Ideal) alongside $\delta = 0$ (Ideal) is grounded in both pedagogical theory and empirical evidence rather than an arbitrary relaxation of the evaluation criterion. Larsen et al. [24] found that the Revised Bloom’s Taxonomy follows a relaxed rather than a strict hierarchy, with overlapping boundaries that make adjacent cognitive levels more difficult to distinguish than distant ones. This observation is supported by Alammary and Masoud [25], who reported that most classification errors in Bloom-level prediction occur between neighboring levels such as Understanding–Applying and Evaluating–Creating. They further noted that these adjacent levels are linguistically and conceptually similar enough to challenge even human annotators. Consequently, a one-level deviation ($\delta = 1$) is considered acceptable, although not ideal, because it remains close to the intended cognitive target. From an evaluation perspective, distinguishing between adjacent and non-adjacent errors enables a more nuanced assessment of system performance, as not all misclassifications carry the same pedagogical significance. Nevertheless, this tolerance is a design choice specific to the proposed pipeline evaluation and may need to be tightened in high-stakes assessment contexts where precise cognitive-level classification is required.

TABLE VII
STATUS CLASSIFICATION BASED ON COGNITIVE DISTANCE

Status	Condition	Interpretation
Ideal	$\delta = 0$	Predicted level exactly matches target level
Slightly Ideal	$\delta = 1$	Predicted level is adjacent to target level
Non-Ideal	$\delta = 2$	Predicted level deviates significantly from target

Questions with status Non-Ideal are rejected and the generation process in Phase 2 is repeated for the same topic τ and target level $L*$. This loop runs for a maximum of three attempts. If an Ideal or Slightly Ideal question is produced before the limit is reached, the loop terminates early and that question is returned as the final output. If all three attempts are exhausted without producing an acceptable result, the question with the smallest δ across all attempts is selected as the final output. The complete result record for each question includes the question text, target level $L*$, predicted level $L^\wedge$, cognitive distance δ, status category, confidence score, and total attempts used.

G. Evaluation Metrics

This study employs three complementary evaluation schemes to assess system performance: classifier evaluation, paper-style evaluation, and efficiency evaluation. The classifier evaluation measures the intrinsic performance of the ModernBERT-Bloom-Taxonomy classifier on the Kaggle Bloom's Taxonomy dataset described in Section 3.2. Four metrics are computed per class using scikit-learn: Accuracy, Precision, Recall, and F1-score. Accuracy measures the proportion of correctly predicted instances out of all instances, as shown in Equation (10).

$$Accuracy = \frac{(TP + TN)}{N} \tag{10}$$

Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive, as shown in Equation (11).

$$Precision = \frac{TP}{(TP + FP)} \tag{11}$$

Recall measures the proportion of correctly predicted positive instances out of all actual positive instances, as shown in Equation (12).

$$Recall = \frac{TP}{(TP + FN)} \tag{12}$$

F1-score is the harmonic mean of Precision and Recall, providing a balanced measure when class distribution is uneven, as shown in Equation (13).

$$F1Score = 2 \times \frac{(Precision\ \times Recall)}{(Precision + Recall)} \tag{13}$$

Per-class metrics are aggregated as weighted averages across all six Bloom's Taxonomy levels, weighted by the number of true instances per class. A confusion matrix is additionally reported to visualize iter-class classification patterns across BT1–BT6.

The paper-style evaluation measures the end-to-end cognitive-level alignment of the full generation pipeline using the cognitive distance metric δ defined in previous section. In this scheme, TP, FP, and FN are redefined based on the value of δ rather than conventional classification correctness. A generated question is counted as a True Positive (TP) when $\delta = 0$, meaning the predicted level exactly matches the target level. A generated question is counted as a False Positive (FP) when $\delta = 1$, meaning the predicted level is one level adjacent to the target. A generated question is counted as a False Negative (FN) when $\delta \geq 2$, meaning the predicted level deviates significantly from the target. Note that the FN category in the paper-style evaluation specifically captures questions that remained Non-Ideal ($\delta \geq 2$) even after the maximum of three regeneration attempts. These are retained in the final output, using their minimum-δ attempt, and are therefore included in all reported metrics. They are not discarded from the dataset. Each topic-level pair therefore contributes exactly one final record to the evaluation, regardless of how many regeneration attempts it required. Using these definitions, Precision, Recall, F1, and Accuracy are computed using the same formulas as Equations (11), (12), (13), and a modified Accuracy as shown in Equation (14). Throughout the remainder of this paper, this end-to-end Accuracy metric is referred to in narrative discussion as the classifier-verified alignment rate, to clearly distinguish it from accuracy validated against independent human judgment. This metric quantifies the proportion of generated questions that the pipeline's own verification classifier deems aligned with the target cognitive level, rather than a measure of pedagogical correctness confirmed by external experts.

$$Accuracy = \frac{TP}{(TP + FP + FN)} \tag{14}$$

Beyond evaluating cognitive-level alignment, this study also measures the computational efficiency of the regenerative feedback mechanism. The evaluation quantifies the time required to complete one retrieval–generation–classification cycle during each regeneration attempt. For the *i*-th attempt, the cycle time, $t_{cycle,i}$, is defined as the sum of the retrieval, question generation, and cognitive-level classification times, as expressed in Equation (15).

$$t_{cycle,i} = t_{retrieval,i} + t_{generate,i} + t_{classify,i} \tag{15}$$

where $t_{retrieval,i}$, $t_{generate,i}$, and $t_{classify,i}$denote the wall-clock time required for Retrieval Augmented Generation (RAG) to retrieve relevant context from the Qdrant vector store, the Large Language Model (LLM) to generate a candidate question, and ModernBERT to classify the cognitive level of the generated question, respectively, during the *i*-th attempt. Because a question may require multiple regeneration attempts before satisfying the target Bloom's Taxonomy level, the reported latency for each completed question is obtained by accumulating the cycle times of all attempts performed by the feedback loop until the question is accepted or the maximum of three attempts is reached.

IV. Results And Discussion

A. Experimental Setup

All experiments were conducted on a local machine equipped with an NVIDIA GeForce RTX 3060 GPU (12 GB

VRAM). The proposed framework was implemented in Python using the following libraries: LangChain for pipeline orchestration, Qdrant (in-memory) as the vector database, HuggingFace Transformers for the ModernBERT-Bloom-Taxonomy classifier, and Ollama for local LLM inference. The complete list of experimental parameters is summarized in Table VIII.

TABLE VIII
EXPERIMENTAL SETUP AND PARAMETERS

Component	Configuration
GPU	NVIDIA GeForce RTX 3060 (12 GB VRAM)
Embedding Model	all-MiniLM-L6-v2 (384-dim, cosine)
Vector Database	Qdrant (in-memory)
Chunk Size	500 characters
Chunk Overlap	50 characters
Top-k Retrieval	3
Classifier	ModernBERT-Bloom-Taxonomy (HuggingFace)
LLM	Llama-3.1-8B, Qwen-3-8B, Gemma-3-4B, Mistral-7B (Ollama, local)
LLM Temperature	0.7
Max Regeneration Attempts	3
Total Topics Evaluated	319
Total Questions Generated	1,914

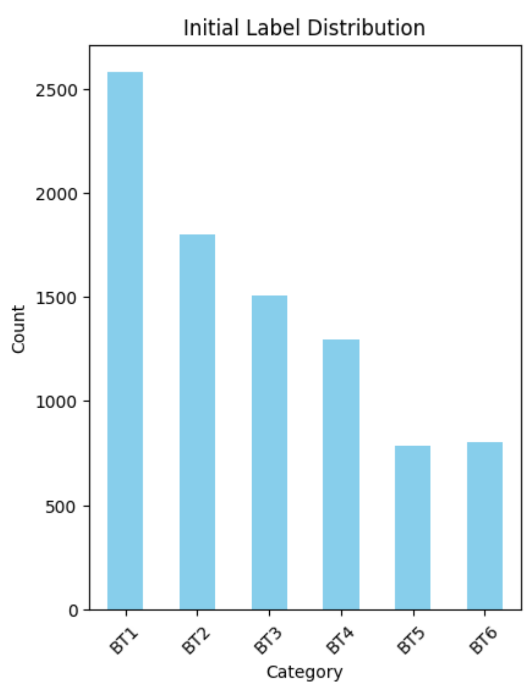

Fig. 3. Label distribution of the bloom's taxonomy classification dataset.

B. Dataset Statistics

This section presents descriptive statistics for both datasets used in the study. The OpenStax knowledge base dataset consists of 350 entries extracted and preprocessed from five subjects, subsequently split into 3,666 chunks for embedding and retrieval. After stratified sampling, the dataset was balanced across the five subjects (Biology, Physics, Chemistry, Psychology, Economics), each contributing approximately 70 entries (350 total).

The Bloom's Taxonomy classification dataset sourced from Kaggle consists of 8,767 labeled questions distributed across six cognitive levels. Figure 3 presents the class distribution of this dataset.

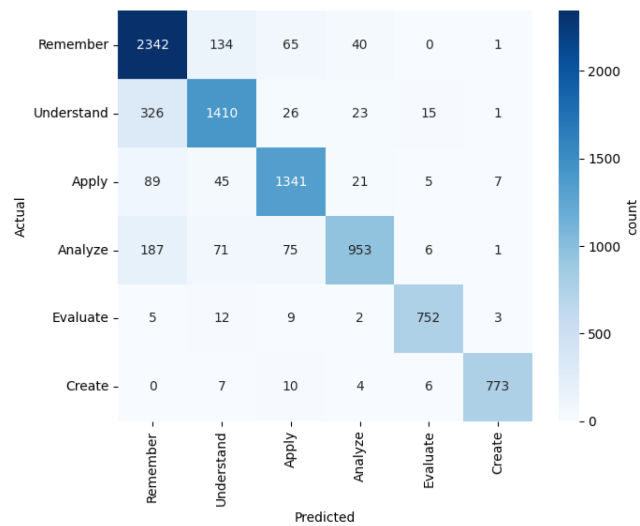

Fig. 4. ModernBert-bloom-taxonomy confusion matrix.

As shown in Figure 4, the dataset exhibits a moderate class imbalance, with Remember (C1) being the most frequent class (29.45%) and Evaluate (C5) being the least frequent (8.93%). This imbalance is consistent with the natural distribution of educational questions, where lower-order cognitive tasks tend to be more prevalent. The weighted average metrics reported in the next section account for this imbalance.

C. Classifier Evaluation Results

The ModernBERT-Bloom-Taxonomy classifier was evaluated on the full Kaggle Bloom's Taxonomy dataset (n = 8,767) to assess its intrinsic performance as the cognitive-level verification component. The overall accuracy achieved was 86.36%, with a weighted precision of 0.8677, weighted recall of 0.8636, and weighted F1-score of 0.8629. The complete per-class results are presented in Table IX.

The results indicate that the classifier performs strongly across most cognitive levels, with particularly high performance on higher-order levels. Create (C6) achieves the highest F1-score of 0.97, followed by Evaluate (C5) at 0.96.

TABLE IX
PER-CLASS CLASSIFIER EVALUATION RESULTS

Bloom's Level	C-Code	Precision	Recall	F1-Score
Remember	C1	0.79	0.91	0.85
Understand	C2	0.84	0.78	0.81
Apply	C3	0.88	0.89	0.88
Analyze	C4	0.91	0.74	0.82
Evaluate	C5	0.96	0.96	0.96
Create	C6	0.98	0.97	0.97
Macro Avg		0.89	0.87	0.88
Weighted Avg		0.87	0.86	0.86
Overall Acc				0.8636

This is consistent with the linguistic characteristics of these levels, as questions targeting C5 and C6 typically contain distinct action verbs such as evaluate, justify, design, and construct that are relatively unambiguous for the classifier.

In contrast, Remember (C1) and Understand (C2) exhibit comparatively lower performance, with F1-scores of 0.85 and 0.81 respectively. The lower precision of Remember (0.79) suggests that the classifier tends to over-predict this class, likely because lower-order questions often use simple and general language that overlaps semantically with Understand-level questions. The lower recall of Analyze (C4) at 0.74 similarly indicates that the classifier frequently misclassifies Analyze-level questions as Understand or Apply, reflecting the linguistic proximity of these adjacent cognitive levels.

These patterns are further confirmed by the confusion matrix (Fig. 4), which shows that the primary source of misclassification is between adjacent levels, particularly between C1 and C2 as well as between C3 and C4. This observation is consistent with the cognitive continuum nature of Bloom's Taxonomy, where neighboring levels share overlapping linguistic features [23], [24], [25], the same property that motivated treating $\delta = 1$ as Slightly Ideal rather than Non-Ideal in the regenerative feedback loop. It should be noted that ModernBERT-Bloom-Taxonomy is a pre-trained model adopted from a publicly available checkpoint; therefore, these results reflect the classifier's intrinsic capability rather than a direct contribution of this study. The reported accuracy of 86.36% confirms that the selected classifier is sufficiently reliable to serve as the cognitive-level verification component within the proposed generation pipeline.

D. Question Generation Results

This section presents the paper-style evaluation results for the end-to-end question generation pipeline across four LLMs: Llama-3.1-8B, Mistral-7B, Qwen-3-8B, and Gemma-3-4B. All models were evaluated under identical experimental conditions using the same RAG pipeline, prompt template, classifier, and feedback loop configuration described in proposed method. A total of 1,914 questions were generated per model (319 topics × 6 Bloom's levels). Table X presents the overall paper-style metrics for each LLM across all Bloom's levels.

Llama-3.1-8B achieves the highest number of Ideal questions (1,549) and the highest overall classifier-verified alignment rate (0.8093), as well as the highest F1-score (0.8946), making it the strongest model overall on the aggregate metrics. Gemma-3-4B follows closely, with the second-highest Ideal count (1,532), alignment rate (0.8004), and F1-score (0.8891). Qwen-3-8B attains the highest Precision (0.8677), indicating fewer Slightly Ideal predictions relative to Ideal ones, but at the cost of a substantially higher Non-Ideal count (296), the largest among all four models, resulting in the lowest Recall (0.8259) and the lowest alignment rate (0.7335). Mistral-7B records the highest Slightly Ideal count (378) among all models and performs at an intermediate level overall, without leading in any single category. Tables XI through XIV present the paper-style metrics broken down per Bloom's Level (BL) for each LLM, including the underlying TP (Ideal), FP (Slightly Ideal), and FN (Non-Ideal) counts that determine each derived metric. Reporting the raw counts in addition to the derived metrics allows the reader to inspect the specific direction in which each model deviates from the target cognitive level: a high FP count indicates predominantly adjacent-level drift, whereas a high FN count indicates substantial cognitive-level deviation. With n = 319 per level (319 topics × one final-attempt question), the three counts always sum to the row total, providing full reproducibility for every derived metric.

Table XV summarizes the F1-score per Bloom's level across all four LLMs for direct comparison. F1-score is used in this context as the primary per-level metric because it provides a balanced measure between Precision and Recall, capturing both the ability of the pipeline to generate Ideal questions and its ability to avoid Non-Ideal outputs simultaneously. A high F1-score at a given level indicates that the LLM consistently produces questions whose predicted cognitive level exactly matches the target level, while a low F1-score reflects either a high rate of Non-Ideal generation or a tendency to produce Slightly Ideal questions at adjacent levels.

The table reveals that, unlike in the previous evaluation run, cognitive-level alignment in the current results is markedly more concentrated in a single model. Llama-3.1-8B achieves the highest F1-score at four of the six levels (C1, C3, C4, C6), making it the most consistently strong model overall, with a perfect F1-score of 1.00 at C6 (Create) and near-perfect performance at C1 (Remember, F1 = 0.98) and C4 (Analyze, F1 = 0.96). Qwen-3-8B leads at C2 (Understand) with an F1-score of 0.99 but collapses at C4 (Analyze) with only 0.48, exhibiting the highest inter-level variance among all models.

Gemma-3-4B is the runner-up at most levels, performing closely behind the leading model at C2, C4, and C6, while Mistral-7B shows moderate performance without leading at any single level. These patterns suggest that the difficulty of cognitive-level generation does not follow a monotonically increasing trend with Bloom's level, but rather reflects the degree of linguistic overlap between adjacent levels,

TABLE X
OVERALL PAPER-STYLE EVALUATION RESULTS PER LLM

LLM	TP (Ideal)	FP (Slightly Ideal)	FN (Non-Ideal)	Total	Precision	Recall	F1-score	Alignment Rate*
Llama-3.1-8B	1,549	346	19	1,914	0.8174	0.9879	0.8946	0.8093
Mistral-7B	1,431	378	105	1,914	0.7910	0.9316	0.8556	0.7476
Qwen-3-8B	1,404	214	296	1,914	0.8677	0.8259	0.8463	0.7335
Gemma-3-4B	1,532	295	87	1,914	0.8385	0.9463	0.8891	0.8004

*Alignment Rate refers to the modified end-to-end Accuracy defined in Equation (14), computed using the ModernBERT classifier's own predictions and not validated against independent human judgment.

TABLE XI
LLAMA-3.1-8B PER-LEVEL PAPER-STYLE RESULTS

BL	TP	FP	FN	Precision	Recall	F1	Alignment Rate
C1	304	12	3	0.96	0.99	0.98	0.95
C2	219	92	8	0.70	0.96	0.81	0.69
C3	270	48	1	0.85	1.00	0.92	0.85
C4	295	22	2	0.93	0.99	0.96	0.92
C5	142	172	5	0.45	0.97	0.62	0.45
C6	319	0	0	1.00	1.00	1.00	1.00

TABLE XII
MISTRAL-7B PER-LEVEL PAPER-STYLE RESULTS

BL	TP	FP	FN	Precision	Recall	F1	Alignment Rate
C1	302	11	6	0.96	0.98	0.97	0.95
C2	241	72	6	0.77	0.98	0.86	0.76
C3	169	146	4	0.54	0.98	0.69	0.53
C4	232	28	59	0.89	0.80	0.84	0.73
C5	202	109	8	0.65	0.96	0.78	0.63
C6	285	12	22	0.96	0.93	0.94	0.89

TABLE XIII
QWEN-3-8B PER-LEVEL PAPER-STYLE RESULTS

BL	TP	FP	FN	Precision	Recall	F1	Alignment Rate
C1	291	11	17	0.96	0.94	0.95	0.91
C2	313	4	2	0.99	0.99	0.99	0.98
C3	206	106	7	0.66	0.97	0.78	0.65
C4	100	26	193	0.79	0.34	0.48	0.31
C5	241	65	13	0.79	0.95	0.86	0.76
C6	253	2	64	0.99	0.80	0.88	0.79

TABLE XIV
GEMMA-3-4B PER-LEVEL PAPER-STYLE RESULTS

BL	TP	FP	FN	Precision	Recall	F1	Alignment Rate
C1	279	13	27	0.96	0.91	0.93	0.87
C2	308	9	2	0.97	0.99	0.98	0.97
C3	157	145	17	0.52	0.90	0.66	0.49
C4	260	33	26	0.89	0.91	0.90	0.82
C5	219	95	5	0.70	0.98	0.81	0.69
C6	309	0	10	1.00	0.97	0.98	0.97

TABLE XV
F1-SCORE PER BLOOM'S LEVEL PER LLM

Bloom's Level	Llama-3.1-8B	Mistral-7B	Qwen-3-8B	Gemma-3-4B
Remember	0.98	0.97	0.95	0.93
Understand	0.81	0.86	0.99	0.98
Apply	0.92	0.69	0.78	0.66
Analyze	0.96	0.84	0.48	0.90
Evaluate	0.62	0.78	0.86	0.81
Create	1.00	0.94	0.88	0.98

particularly between C2 and C3, and between C4 and C5, which varies in sensitivity across different LLM architectures.

E. Efficiency And Scalability Analysis

Beyond cognitive-level alignment, a central claim of this study is that the proposed pipeline supports scalable automated question generation. To substantiate this claim with empirical evidence rather than relying solely on the total number of generated questions, this section evaluates the latency, regeneration behavior, and throughput of the complete retrieval, generation, and classification cycle measured during the 1,914-question evaluation run for each Large Language Model.

Table XVI presents the latency breakdown of the retrieval,

TABLE XVI
LATENCY BREAKDOWN PER LLM FOR THE RETRIEVAL, GENERATION, AND CLASSIFICATION CYCLE

Model	Retrieval (s)	LLM Generation (s)	Classifier (s)	Total Cycle (s)	Median Cycle (s)	P95 Cycle (s)
Llama-3.1-8B	0.0124	0.9966	0.0199	1.0289	0.8405	2.1229
Mistral-7B	0.0138	2.6781	0.0247	2.7166	2.1722	6.3784
Qwen-3-8B	0.0155	1.1892	0.0267	1.2315	0.9097	2.7828
Gemma-3-4B	0.0121	1.4126	0.0222	1.4469	1.2446	3.1914

generation, and classification cycle, averaged across all completed questions. The reported cycle latency includes the cumulative processing time of all regeneration attempts required for each question before it satisfied the target cognitive level or reached the maximum of three attempts. Across all four models, LLM generation accounts for the largest proportion of the total latency, whereas retrieval from the Qdrant vector store and cognitive-level classification using ModernBERT contribute only a small fraction of the overall processing time. Llama-3.1-8B achieves the lowest mean cycle time of 1.03 s, while Mistral-7B records the highest mean cycle time of 2.72 s, reflecting differences in decoding efficiency under the same 4-bit quantization setting.

TABLE XVII
DISTRIBUTION OF REGENERATION ATTEMPTS PER LLM

Model	1× (%)	2× (%)	3× (%)	Mean Attempts
Llama-3.1-8B	87.51	9.30	3.19	1.16
Mistral-7B	77.48	12.90	9.61	1.32
Qwen-3-8B	76.75	5.12	18.13	1.41
Gemma-3-4B	87.30	5.90	6.79	1.19

TABLE XVIII
TOTAL WALL CLOCK TIME AND THROUGHPUT FOR QUESTION EVALUATION

Model	Total Wall-Clock Time	Throughput (q/s)	Throughput (q/hour)
Llama-3.1-8B	1,969.4 s	0.9719	3,498.8
Mistral-7B	5,199.5 s	0.3681	1,325.2
Qwen-3-8B	2,357.1 s	0.8120	2,923.3
Gemma-3-4B	2,769.4 s	0.6911	2,488.0

Table XVII summarizes the number of regeneration attempts required before each generated question satisfied the target cognitive level. Llama-3.1-8B and Gemma-3-4B successfully completed most questions on the first attempt, accounting for 87.51% and 87.30% of all generated questions, respectively, and consequently achieved the lowest mean number of attempts. In contrast, Qwen-3-8B required the highest average number of attempts and produced the largest proportion of questions that reached the maximum of three attempts.

Table XVIII reports the overall processing time and throughput achieved by the complete pipeline for each evaluated model. Based on the measured wall-clock time of the 1,914-question evaluation run, the pipeline achieves a throughput ranging from 0.37 to 0.97 questions per second, corresponding to approximately 1,325 to 3,499 cognitively verified questions per hour without batching or parallel processing. Under these conditions, completing the full evaluation set requires between 32.8 minutes for Llama-3.1-8B and 86.7 minutes for Mistral-7B. These results support the scalability claim presented in the Abstract and Introduction by demonstrating that a single local deployment can generate thousands of cognitively verified questions within a few hours, making large-scale question bank construction practical for institutional educational settings. Throughput could be further improved through batched inference or multi-GPU deployment.

F. Discussion

Llama-3.1-8B and Gemma-3-4B emerge as the two strongest models overall, with alignment rates of 0.8093 and 0.8004 and F1-scores of 0.8946 and 0.8891 respectively. Llama-3.1-8B leads on both metrics, reflecting its ability to produce the highest proportion of exact cognitive-level matches (Ideal), with 1,549 out of 1,914 questions, while keeping the Non-Ideal count to only 19, the lowest among all four models. The performance gap between these two leading models and the remaining two is moderate in F1-score but more pronounced in alignment rate and in the FN (Non-Ideal) count. This shows that Llama-3.1-8B and Gemma-3-4B not only approximate the target cognitive level more often, but also rarely generate questions that are completely off-target.

Qwen-3-8B achieves the highest Precision (0.8677) among all models, indicating that when it does produce an Ideal question, it is rarely accompanied by Slightly Ideal outputs. However, its substantially high Non-Ideal count (296), more than fifteen times that of Llama-3.1-8B (19), suggests that Qwen-3-8B tends to either produce exactly correct or significantly off-target questions, with less tolerance for adjacent-level generation. This behavior is most critically observed at C4 (Analyze), where Qwen-3-8B achieves an F1-score of only 0.48 with 193 Non-Ideal questions, representing the worst single-level result across all models. This suggests that Qwen-3-8B struggles to internalize the linguistic distinction between Analyze-level and lower-order cognitive tasks within the prompt-guided generation setting. In contrast, Mistral-7B shows moderate and relatively balanced performance across most levels, with its primary weakness at C3 (Apply), where a high FP count of 146 indicates that many Apply-level questions are generated at an adjacent level. This suggests that Mistral-7B frequently interprets Apply-level instructions in the prompt as Understand-level tasks.

Examining individual Bloom's levels reveals consistent patterns across the four models. C1 (Remember) is consistently the highest-performing level across all models, with F1-scores ranging from 0.93 (Gemma-3-4B) to 0.98 (Llama-3.1-8B). This is expected, as Remember-level questions typically use simple and unambiguous recall-based language that is straightforwardly distinguishable by both the LLM and the classifier. This result is consistent with prior studies that characterize Remember as the lowest cognitive level in Bloom's Taxonomy, requiring primarily factual recall and recognition rather than higher-order reasoning processes [26].

At C2 (Understand), the results show high variability across models. Qwen-3-8B and Gemma-3-4B achieve F1-scores of 0.99 and 0.98 respectively, whereas Llama-3.1-8B achieves only 0.81, the lowest among the four. The lower Llama performance at this level is attributable to a high FP count (92), indicating that Llama-3.1-8B frequently generates questions that the classifier predicts as C1 (Remember) rather than C2 (Understand). This is a consistent confusion pattern also observed in the classifier evaluation.

C3 (Apply) is the most problematic level for three of the four models. Mistral-7B and Gemma-3-4B achieve F1-scores of 0.69 and 0.66 respectively, while Qwen-3-8B achieves 0.78. Only Llama-3.1-8B achieves a substantially higher F1-score of 0.92 at this level. The high FP counts across the other three models suggest that Apply-level prompts tend to elicit questions that are classified as Understand-level, reflecting the linguistic overlap between these two adjacent cognitive levels. This confusion is consistent with prior Bloom's Taxonomy studies reporting substantial semantic overlap between Understand and Apply categories, where questions often differ only in the degree of contextual application rather than linguistic form [23], [24], [25].

C4 (Analyze) exhibits the most dramatic inter-model disparity. Llama-3.1-8B achieves 0.96 and Gemma-3-4B achieves 0.90, while Qwen-3-8B collapses to 0.48 with 193 Non-Ideal results. This extreme failure of Qwen-3-8B at C4 suggests a systematic tendency to generate questions at substantially lower cognitive levels when targeting Analyze, possibly due to differences in how the model interprets analytical action verbs such as analyze, contrast, and differentiate within the prompt context. Prior studies have reported that Analyze-level questions are among the most difficult categories to identify automatically because analytical reasoning is often expressed through linguistic patterns that overlap with lower cognitive levels, resulting in frequent misclassification and generation drift [23] [27].

C5 (Evaluate) is the weakest level for Llama-3.1-8B (F1 = 0.62), driven primarily by a very high FP count of 172. This means that the majority of generated questions classified as non-Ideal at this level are Slightly Ideal rather than completely off-target. This indicates that Llama-3.1-8B tends to generate Evaluate-level questions that the classifier places at C4 (Analyze) or C6 (Create), suggesting proximity between the LLM's interpretation of Evaluate and adjacent higher-order levels. By contrast, Mistral-7B (0.78), Qwen-3-8B (0.86), and Gemma-3-4B (0.81) all perform notably better at C5, with Qwen-3-8B achieving the highest F1-score at this level.

C6 (Create) is consistently strong across all models, with Llama-3.1-8B achieving a perfect F1-score of 1.00, followed by Gemma-3-4B at 0.98. Llama-3.1-8B achieves a perfect Precision of 1.00 and a perfect Recall of 1.00 at C6, meaning that all 319 of its generated Create-level questions at this level were classified as exactly Ideal, with zero Slightly Ideal or Non-Ideal outputs.

Taken together, Llama-3.1-8B demonstrates the most balanced and strongest performance profile, achieving the highest F1-score at four out of six Bloom's levels (C1, C3, C4, C6), the best overall F1-score of 0.8946, and the highest overall alignment rate of 0.8093. Its primary weakness lies at C5 (Evaluate), where a large proportion of generated questions are classified as adjacent levels. Gemma-3-4B follows closely as the second-best model overall, with strong performance across most levels, particularly at C2 and C6. Qwen-3-8B excels at C2 but exhibits a critical failure at C4, making it unsuitable as a standalone generation model without additional prompt tuning. Mistral-7B provides moderate and consistent performance but does not lead in any single metric, suggesting it is the least effective model within this pipeline for cognitive-level question generation. These results confirm that the choice of LLM has a significant impact on cognitive-level alignment quality. They also confirm that Llama-3.1-8B represents the most suitable model for the proposed RAG-based Bloom's Taxonomy question generation framework under the current experimental conditions.

Beyond cognitive-level alignment, the practical applicability of the proposed framework also depends on its computational efficiency and scalability. The efficiency evaluation demonstrates that the proposed framework is computationally practical for large-scale automated question generation. Although the regenerative feedback mechanism introduces additional processing cycles when a generated question does not satisfy the target cognitive level, most questions produced by Llama-3.1-8B and Gemma-3-4B are successfully completed on the first attempt. Consequently, these models achieve lower average cycle latency while sustaining higher end-to-end throughput than the remaining models. These findings indicate that classifier-based verification and regenerative feedback improve cognitive-level alignment without imposing a substantial computational overhead on the complete pipeline.

TABLE XIX
COMPARISON OF PROPOSED FRAMEWORK RESULTS AGAINST PRIOR WORK

Aspect	Aisyah et al. [5]	Banujan et al. [6]	Scaria et al. [12]	Proposed Work
Question Generation	✓	✗	✓	✓
Bloom's Taxonomy (C1–C6)	✗	✓	✓ (prompt-guided)	✓ (classifier-verified)
Cognitive Verification	✗	✓ (existing questions)	✗	✓ (generated questions)
Feedback Loop	✗	✗	✗	✓
RAG-based Retrieval	✓	✗	✗	✓
Multi-LLM Comparison	✓	✗	✓	✓
Classifier Accuracy	N/A	84.2% (LSTM+BERT)	N/A	86.36% (ModernBERT)
Cognitive Alignment Metric	None	N/A	Qualitative	Ideal Rate 78.84–79.26%
Best Performing Model	Llama-3-8B	LSTM+BERT	GPT-4	Llama-3.1-8B

The regeneration analysis further demonstrates a close relationship between generation quality and computational efficiency. Models that generate cognitively appropriate questions more consistently require fewer regeneration attempts, thereby reducing the cumulative retrieval, generation, and classification time required to complete each question. This relationship is reflected in the throughput results, where the proposed framework produces between 1,325 and 3,499 cognitively verified questions per hour under the experimental configuration used in this study. These findings suggest that the framework is suitable not only for improving cognitive-level alignment but also for supporting the practical construction of large-scale educational question banks.

The results of the proposed framework can be contextualized against prior work reviewed in related work section. Table XIX. summarizes the key distinctions between this study and the most closely related prior studies across several capability dimensions.

G. Limitations

A key limitation of this study concerns the circularity inherent in the end-to-end evaluation. The same ModernBERT-Bloom-Taxonomy classifier that governs the regenerative feedback loop in Phase 3 is also used to define the ground truth against which the reported Precision, Recall, F1-score, and Accuracy (classifier-verified alignment rate) are computed. Consequently, these metrics quantify the degree to which the generation pipeline conforms to its own internal verification signal, rather than pedagogical correctness confirmed by an independent, external authority such as a human domain expert. A generated question classified as Ideal is therefore guaranteed only to align with ModernBERT's interpretation of the target Bloom's level, not necessarily with the judgment of a human educator. This distinction does not invalidate the reported results, since the classifier itself achieves 86.36% accuracy against an independently labeled benchmark dataset (Section IV.C), providing a degree of external grounding for its reliability. However, the end-to-end metrics in Section IV.D should be interpreted as a measure of self-consistency between the generator and the verifier, not as a direct substitute for human expert validation of pedagogical quality. Closing this gap with independent human annotation, ideally through stratified sampling across Bloom's levels and inter-rater agreement analysis such as Cohen's or Fleiss' kappa, is identified as a priority direction for future work.

V. CONCLUSION

This study addressed the limitation of existing automatic question generation approaches that cannot guarantee alignment with specific cognitive levels of Bloom's Taxonomy. To overcome this limitation, a unified framework integrating Retrieval Augmented Generation, Large Language Models, and a pretrained ModernBERT Bloom Taxonomy classifier was proposed. The framework combines contextual retrieval, prompt-guided question generation, classifier-based cognitive verification, and a regenerative feedback loop to generate contextually relevant questions aligned with the six cognitive levels of the Revised Bloom's Taxonomy, replacing the coarse low, medium, and high difficulty categories commonly adopted in earlier RAG-based studies.

Experimental evaluation across four representative open-weight Large Language Models demonstrates the effectiveness of the proposed framework. The pretrained ModernBERT classifier proved sufficiently reliable to serve as the cognitive verification component within the pipeline. Among the evaluated models, Llama-3.1-8B achieved the strongest overall performance and led at the majority of Bloom's levels, while Gemma-3-4B followed closely as the second-best model. Qwen-3-8B attained the highest precision but struggled at the Analyze level, whereas Mistral-7B delivered more balanced but comparatively lower overall performance. The regenerative feedback mechanism further improved cognitive-level alignment by iteratively refining questions that initially failed verification.

Beyond cognitive-level alignment, the proposed framework also demonstrated practical computational efficiency for large-scale automated question generation. The complete pipeline sustained a throughput suitable for generating large volumes of cognitively verified questions within a short evaluation window, indicating that classifier-based verification and regenerative feedback can be incorporated without substantially limiting scalability. These findings demonstrate that the proposed framework provides an effective and scalable solution for generating pedagogically meaningful assessment items, making it suitable for educational applications such as large-scale question bank construction and adaptive assessment within Learning Management Systems.

The analysis also showed that cognitive-level alignment is influenced by both the semantic overlap between adjacent Bloom's levels and the instruction-following behavior of individual Large Language Models. The most consistent confusion occurred between adjacent cognitive levels, suggesting that model selection should consider the target distribution of cognitive complexity required in a particular educational context. Rather than replacing human experts, the proposed framework is intended to function as an assistive tool that supports educators by efficiently generating assessment items with verifiable cognitive levels. Future work will prioritize independent human expert validation, comparing educator-assigned Bloom's-level judgments against the classifier's predictions on a stratified sample of generated questions, to directly address the self-referential nature of the current end-to-end evaluation. Beyond this, the framework will be extended to multilingual educational settings, improve cognitive verification through domain-specific fine-tuning of the ModernBERT classifier and more advanced feedback optimization techniques, and evaluate the framework across broader knowledge domains. These directions are expected to further enhance the robustness, generalizability, and practical applicability of the proposed framework.

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